

HistoColAI User Guide V1



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1 Introduction and log in

HistoColAi is a web platform for visualization, annotation and automatic evaluation of histological cases that supports the identification of different types of cancer. This tool allows pathologists from all over the world to obtain an online diagnostic aid based on artificial intelligence techniques.

The first view of the application is the log in page.

Welcome to HistoColAi the Annotation Tool



Log in

Name
Password

Get in
[Register](#)

FIGURE 1. LOG IN VIEW

On this page you can log in or register a new user, in case of willing of registration you must press "Register" link, in that case you will drive to the following view

Welcome to HistoColAi the Annotation Tool



Log in

Name
E-Mail
Password
Verify password

Send Cancel

FIGURE 2: REGISTER VIEW

In case of the registration have been processed correctly, you will be drive to the main page:

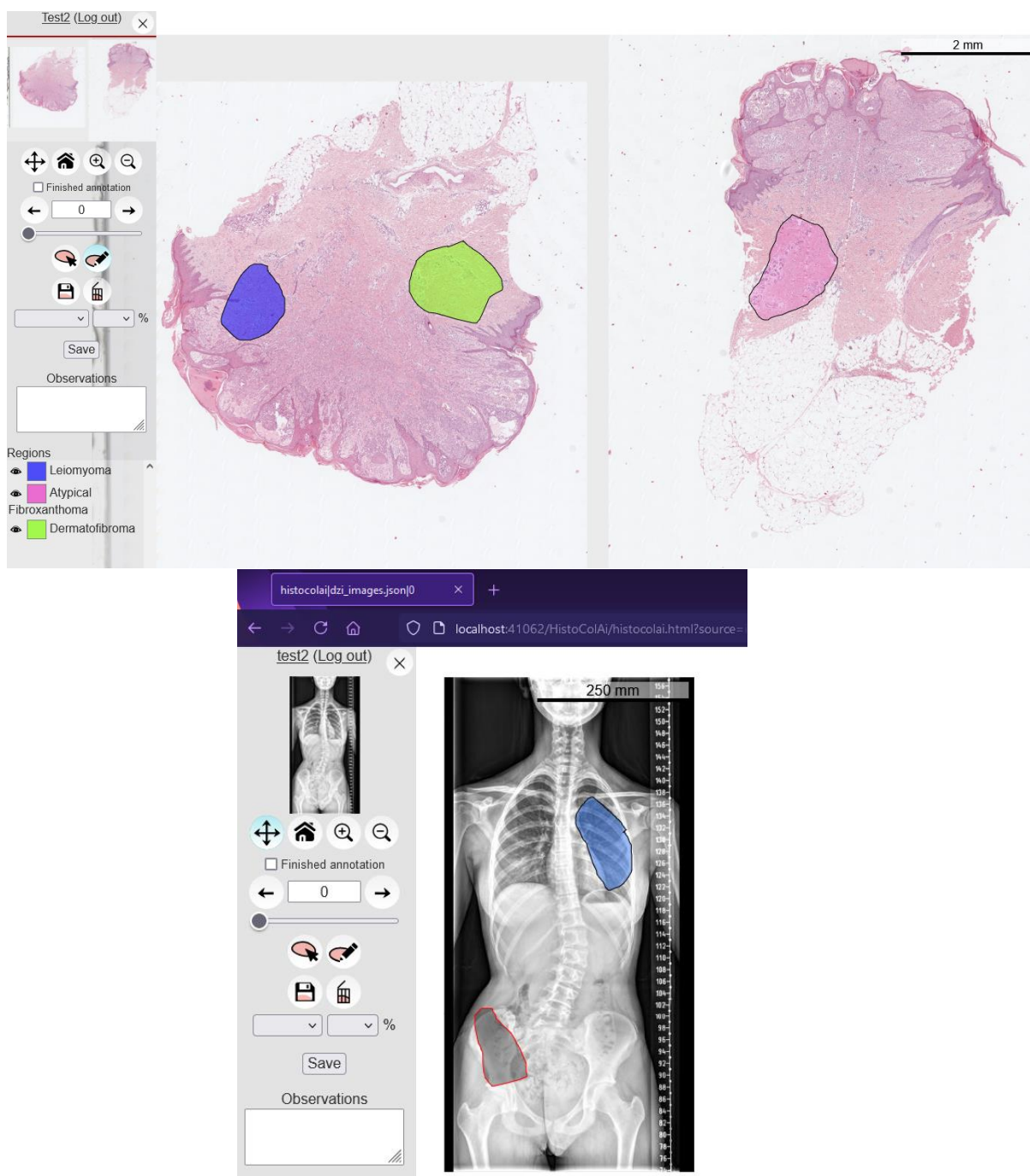


FIGURE 3: MAIN VIEW

2 Annotation Tool

This tool is implemented using php, which runs on the server side (queries the database) and then returns a page in HTML and Javascript. The database is MariaDB, a free database based on MySQL. Javascript libraries such as OpenSeaDragon and PaperJS are used for displaying images and annotations.

2.1 User authentication functionality

Authentication is a process by which you verify that users are who they say they are and have the proper permissions to access different functionalities. In this case, users are maintained and created in the system's own database, and when they login their session is maintained by PHP session variables, so you know which user is making the requests.

2.2 Annotation functionality

Users will annotate the slices of each batch.

Access to batches is not controlled, all users can access as long as they have the URL address for that particular batch.

For annotation, the panel tools will be used:

- **Navigate** (): By clicking here and then dragging on the image, the image is scrolled.
- **Go Home** (): Returns to the original zoom.
- **Zoom In** (): Zoom in, this can be achieved with the mouse wheel as well.
- **Zoom out** (): Zooming out, can also be done with the mouse wheel.
- **Slice change**: It can be changed with the arrows and a slider to change slice sequentially, also with the cursors. In addition, you can type the name of the slice directly if you know.

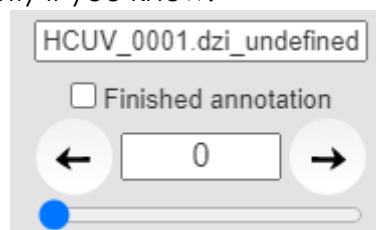


FIGURA 4. SLICE CHANGE

- **Select** (): When clicked, you can then select annotations in the image. This can also be achieved by clicking on the name of the annotation in the list.
- **Draw** (): When selected and clicking on an area of the image will start to draw an area where the mouse passes, until you stop clicking.
- **Save** (): All annotations and changes that have been made are saved. There is a process that saves every 5 minutes so that the data is not lost.
- **Delete** (): The selected region is deleted. This can also be done by pressing the DELETE key.
- **Undo**: There is no button, but pressing CTRL+Z undoes the last action.
- **Redo**: There is no button, but pressing CTRL+Y redoes the last undo.
- **Copy**: Having a region selected, and pressing CTRL+C, it is copied.
- **Paste**: The previously copied region is pasted by pressing CTRL+V.
- **Chose label percentage**: There are drop-downs for the scorer to choose the label and percentage.

Leiomyom ▾	10 ▾	%
Leiomyosar ▾	40 ▾	%
Dermatofit ▾	50 ▾	%

FIGURA 5. LABELS AND PERCENTAGES

- **List of regions**: A list with all annotated regions, if selected, this region will be marked on the image. If you click on the eye drawing, that region will be shown or hidden in the image. If you double click on a region in the list, the list of labels will appear so that you can change the label of that region.

Regions	
	 Malignant
	 Benign
	 Healthy

FIGURA 6. LIST OF REGIONS.



FIGURA 7. LIST OF TAGS.